NATIONAL JUNIOR COLLEGE
2008 TERM 4 PRELIMINARY EXAMINATION

MATHEMATICS **9810**
Higher 3

Additional Materials: Answer Paper
List of Formulae (MF15)
Cover Sheet

3 hours

10^{th} September 2008 Wednesday 1400 – 1700 hr

INSTRUCTIONS TO CANDIDATES

Write your name and registration number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** the questions in Section A and any **four** questions from Section B.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers. Up to 4 marks will be awarded for the style and clarity of your mathematical presentation, based on the whole paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

This question booklet consists of 8 printed pages

[Turn over

Section A

Answer **all** the questions in this section.

1 Consider the following differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y^2 - 2y + k,$$

where k is a parameter and $k \in \mathbb{R}$.

Sketch a labelled bifurcation diagram as the parameter k is varied in the interval $[-1, 3]$. [5]

2 A researcher wish to model a population with a differential equation of the form $\frac{\mathrm{d}P}{\mathrm{d}t} = \mathrm{f}(P)$, where $P(t)$ is the population at time t. Experiments have been performed on the population and the following information are obtained:

1. The only equilibrium points in the population are $P = 0$, $P = 20$ and $P = 60$.
2. If the population is 80, the population decreases.
3. If the population is 40, the population will increase.

(i) Determine the nature of equilibrium point at $P = 60$. [1]

(ii) Sketch two possible phase lines for this system for $P > 0$ and give a sketch of their corresponding solution curves. [5]

3 A fish farmer owns a large fishing pond. The population $P(t)$ of fish in the pond is governed by the model $\frac{\mathrm{d}P}{\mathrm{d}t} = 3P - \frac{3P^2}{200}$, where t is time in weeks.

(i) Determine the carrying capacity of the pond. [1]

The owner of the pond decides to let people fish in his pond once a week, but insists that only a certain number of fishes may be removed each time.

(ii) What is the maximum number of fishes that may be harvested each week to ensure the continual survival of the fish population? [3]

(iii) Suppose the initial fish population is $P_0 = 200$, discuss the long tem impacts on the fish population based on the harvesting rate in (ii). Justify your conclusion. [2]

[Turn over

4 Consider the initial value problem

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{4\mathrm{e}^{y}}{\mathrm{e}^{2y}-1}, \quad y(1)=\alpha \text{ where } \alpha > 1.$$

(i) Using the Euler's method with *one* iteration, it was found that the estimated value of y when $x = 1.5$ is approximately 1.89432.

Find the value of α correct to 3 decimal places. [2]

(ii) By using the value of α found in **(i)**, apply the improved Euler's method with a step size of 0.25 to estimate the value of $y(1.5)$, giving your answer corrected to 4 decimal places. [3]

(iii) Given that the exact value of $y(1.5)$ is 1.7627, compute the percentage error in **(i)** and **(ii)**. [2]

(iv) Suggest a method of improving the iterated answer found in **(ii)**. [1]

(v) Describe the nature of the gradient for the graph $y = \mathrm{f}(x)$ as $y \to 0$ and $y \to \infty$.

Hence, give a sketch of the graph $y = \mathrm{f}(x)$ for $x > 0$. [3]

5 Two biological cultures, X and Y, react with each other, and their volumes at time t are x and y respectively, in appropriate units. Their rates of growth are modeled by the equations

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= (2-x)y \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= \frac{y^2}{x} \end{aligned}.$$

(i) When $t=0$, $x = y = 1$. Find x in terms of y. [4]

(ii) Find and simplify expressions for y and x in terms of t. [6]

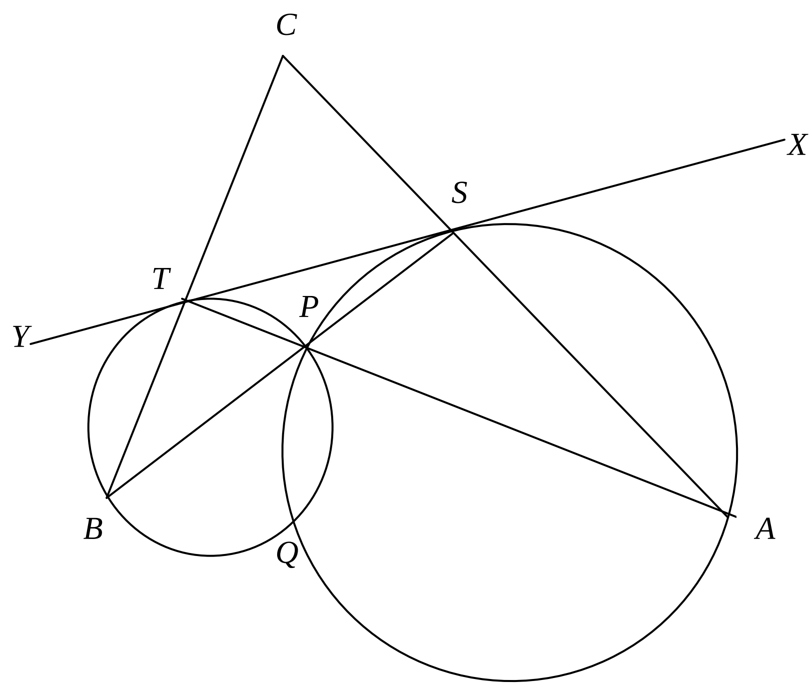
(iii) Sketch the graph of y against x for $0 < t < \frac{1}{2}\pi$. [2]

Section B

Answer any **four** questions from this section. Each question in this section carries 14 marks.

Plane Geometry

6



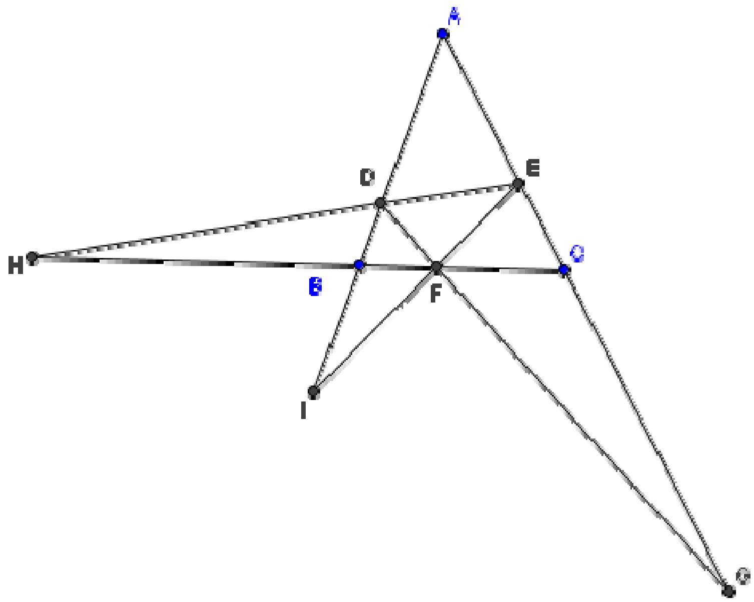
The diagram shows two intersecting circles. The points A, S, P and Q lie on one circle. The points B, T, P and Q lie on the other circle. The line $XSTY$ is a common tangent to the two circles. Line AT and BS intersect at P. Lines AS and BT intersect at C.

(i) Show that triangle STC is isosceles. [5]

(ii) Show that if the points A, Q and B are collinear, then triangle STC is equilateral. [6]

(iii) Given that $PT = 1$ and $PS = 3$, show that $PA - 3PB = 8$. [3]

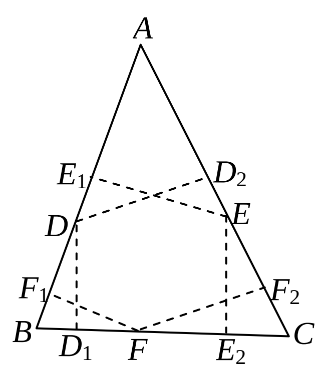
7 **(a)** Triangle DEF is the orthotriangle in triangle ABC as shown in the diagram below. DF is extended to meet AC at G; ED is extended to meet CB at H and EF are extended to meet AB at I.



(i) Show that the ratio $\dfrac{AG}{GC} = -\dfrac{AE}{EC}$. [3]

(ii) Show that G, I and H are collinear. [5]

(b) Triangle DEF is the orthotriangle in triangle ABC as shown in the diagram below. In triangle ABC, D_1 and D_2 are the feet of the perpendiculars from D to BC and AC respectively. Similarly, E_1 and E_2 are the feet of the perpendiculars from E to AB and BC respectively and F_1 and F_2 are the feet of the perpendiculars from F to AB and AC respectively . Let H be the orthocenter of triangle ABC.



(i) Prove that $\dfrac{BH}{BE} = \dfrac{BF}{BE_2}$. [2]

(ii) Show triangle BDD_1 is similar to triangle BFF_1. [1]

(iii) Show that (BF_1) (BE_1) = (BD_1) (BE_2). Deduce that $D_1F_1E_1E_2$ forms a cyclic quadrilateral. [3]

[Turn over

Graph Theory

8 **(a)** An edge e in a connected graph G is called a bridge if $G - e$ is disconnected.

(i) Show that e cannot be contained in any cycle in G. [3]

(ii) If G is a tree, show that every edge is a bridge. [4]

(iii) Does G contain any bridges if G is an Eulerian graph? Justify your answer. [2]

(b) A simple graph has 20 vertices. Any two distinct vertices u and v are such that $\deg(u) + \deg(v) \geq 19$. Prove that the graph is connected. [5]

9 **(a)** Suppose that G is a simple connected graph with 7 vertices and 19 edges.

(i) Prove that the degree of every vertex is at least 4. [3]

(ii) Deduce if G is Hamiltonian. Justify your answer [2]

(iii) Seven people are to be seated around a round table. Each of the seven people mutually knows at least four other people. Is it possible to seat the seven people so that everyone knows the people seated on either side of them? Explain your answer. [3]

(b) There are twelve workers on a building site. On one particular day, ten different jobs have to be completed. Each of the twelve workers is qualified to perform exactly three of these ten jobs.

If each job is such that at least three workers are qualified to perform it, is it possible to assign workers to jobs so that each of the ten jobs is being carried out at the same time? Justify your answer. [6]

[Turn over

Combinatorics

10 **(a)** The Stirling number of the second kind, denoted by $S(r,n)$, is defined as the number of ways of distributing r distinct objects into n identical boxes such that **no box is empty**. Using a combinatorial argument, show that

(i) $S(r,2)=2^{r-1}-1$ [3]

(ii) $S(r,r-2)=\binom{r}{3}+3\binom{r}{4}$ [3]

(b) Mr. Yap brought his ten H3 Mathematics students for a movie. At the theatre, they decided to buy some food from the snack bar which sells sushi, ice-cream, hotdog, nachos and chocolate bars.

(i) If all the students buy a snack each, find the number of ways this can be done. [1]

(ii) Some students bought the family pack of ten items. How many different types of family packs could they choose from if they must have at least three chocolate bars and not more than two hotdogs in the pack. [3]

(iii) There is a super deal promotion for eight scoops of same flavour ice-cream. Suppose each cone can hold a maximum of three scoops, in how many ways can the attendant distribute eight scoops of ice-cream? Suppose five boys decide to order the promotion to share eight scoops of ice-cream, how many ways can this be done (You may assume that each boy has only one cone)? [4]

11 Let C_n denote the number of ways of arranging the numbers 1, 2, 3, …, n such that k is not followed immediately by k + 1 for all values of k = 1, 2, 3, …, n – 1, (i.e. there are no consecutive pairs in the arrangement.) By considering Principle of Inclusion and Exclusion or otherwise, show that $C_n=\sum_{i=0}^{n-1}(-1)^i\binom{n-1}{i}(n-i)!$. [6]

Let D_n denote the number of permutation of the numbers 1, 2, 3, …, n such that the number i is not in the i th position for all i = 1, 2, 3, …, n, i.e. (the positions of all the numbers are changed). Explain clearly the recurrence relation $D_n=(n-1)(D_{n-1}+D_{n-2})$, $n\geq 3$. [4]

Find D_2,D_3,D_4,D_5 and C_1,C_2,C_3,C_4 and conjecture a relation between the two sequences. (You do not need to prove the relation.) [4]

******* THE END *******

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Write your name and registration number in the spaces provided on the cover sheet and attached it on top of your answer paper.

Circle the questions you have attempted and arrange your answers in **NUMERICAL ORDER**.

Name:____________________________ Registration Number:_______________

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<tr><th>Question</th><th>Marks</th><th>Max. Marks</th><th>Question</th><th>Marks</th><th>Max. Marks</th></tr>
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